Optimizers: Steering NNs Towards Better Learning

Thursday, 8 October 2026 7:04 PM

Optimisers :

Why do we need Optimisers?

Our models have some weights = $[w_1\ w_2\ w_3 \dots]$

After prediction, we calculate the Loss.

→ The goal of model training is to find weights that makes the loss as small as possible.

Backprop gives us → $\frac{\partial L}{\partial W}$

↓

If I change the weights, how will the loss change.

$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W}$ → gradient.

step size.

→ The optimiser is the algorithm that decides how to use these gradients to update the weights.

1B data points

1000 mini batches

1B/1000

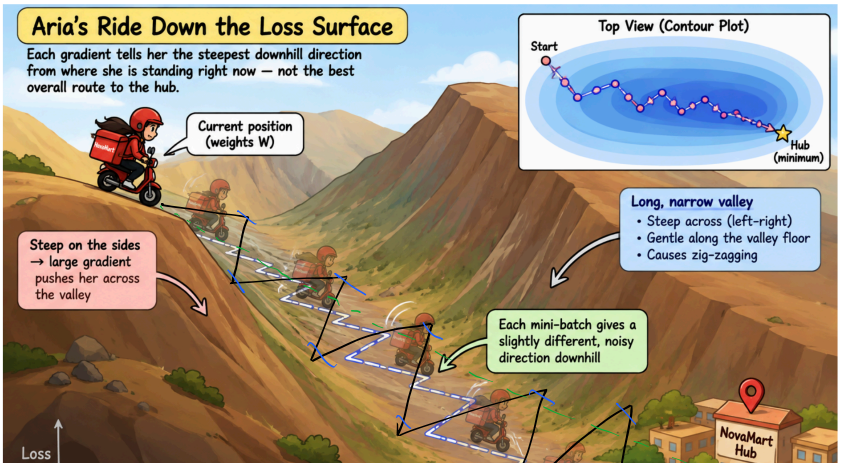
* Gradient Descent

1) Vanilla GD — all 1000 data points

2) Mini Batch GD — mini batch, where each batch is < 1000.

3) SGD — 1 data point.

Let's say we have n = 1000 data points.



✳ Momentum.

It says: Instead of looking only at the current gradient, can I remember where I was going previously.

Moving Average.

Maintain a running average of gradients.

$$V_t = \beta \cdot V_{t-1} + (1-\beta)\, g_t$$

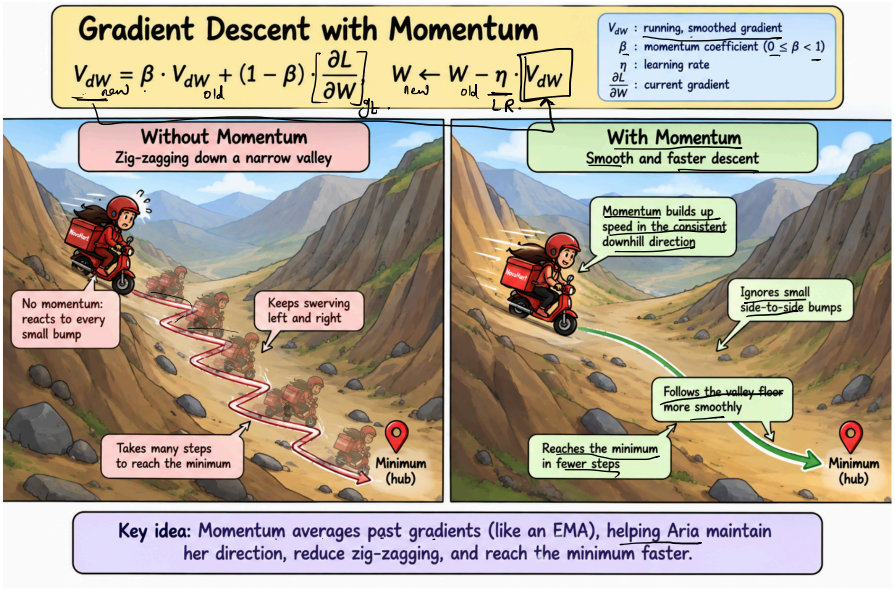
g_t = current gradient $(\partial L/\partial W)$

V_t = Smoothed direction.

β = how much history to retain. $[0-1]$

η = learning rate.

✳ GD with Momentum.



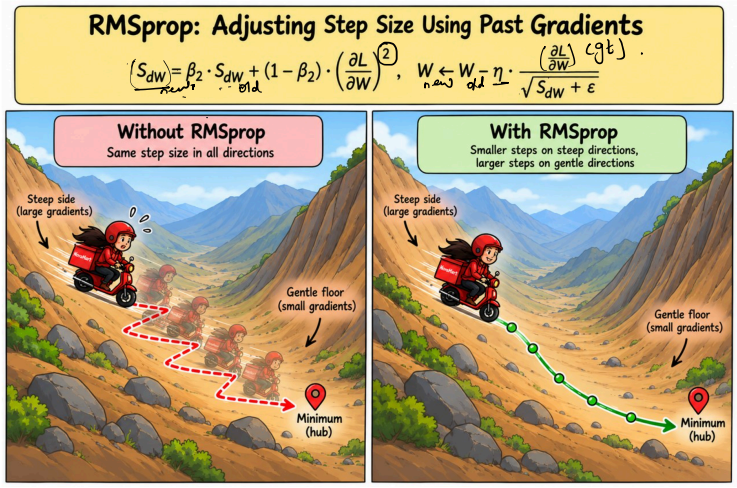
Momentum : How much of Aria's previous direction carries over to the next step?

Can I remember the direction?

✳ Root Mean Square Propogation.

It tracks the moving average of squared gradients. Because squaring

removes sign, it magnifies the recent magnitude of J.



$S_t = 100$.

$\sqrt{S_t} = 10$

g_t 10 → 1 → Smaller.

$S_t = 1$.

g_t 10 → 10. → larger.

Why Square?

+5
-5
+5
-5

If we calculate simple avg → $\frac{+5+(-5)+5+(-5)}{} = 0$

gradient is very small.

we capture the magnitude.

It says, Larger historical gradients, smaller steps.

Smaller historical gradients, larger steps.

✳ Adam. (Adaptive Moment Estimation).

Momentum + RMS prop.

1) Momentum

$V_t = \beta_1 V_{t-1} + (1-\beta_1) g_t$.

This gives direction.

2) RMSProp:

$$S_t = \beta_2 S_{t-1} + (1-\beta_2) g_t^2$$

S_t : numbers magnitude.

Bias Correction:

$$\hat{V}_t = \frac{V_t}{1-\beta_1}$$

$$\hat{S}_t = \frac{S_t}{1-\beta_2}$$

$$W_{new} = W_{old} - \eta \cdot \frac{\hat{V}_t}{\sqrt{\hat{S}_t + \epsilon}}$$

$\hat{V}_t$: momentum.

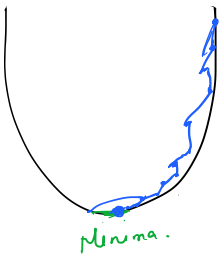
ϵ : ignore division by zero.

RMSProp.

* Learning Rate Decay.

Step Size.

Q. Instead of a constant step size, can we have a step size which decreases with epochs?



$$\eta_{new} = \frac{\eta_0}{1 + r_0 \cdot epoch}$$

(r_0 = 0.05)

Epoch 1

$$\frac{0.01}{1 + 0.05 \times 1} \rightarrow \frac{0.01}{1.05}$$

Epoch 2:

$$\frac{0.01}{1.1}$$

Epochs

0.01

1.15

0.01

large value.

LR

Shrinks.